Lab 16: Capstone Systems Synthesis

BIOL-1

Name: ______________________ **Date:** ______________________

Learning Objectives

By the end of this lab, you should be able to:

1. Connect cell structure, genetics, evolution, and ecology in one causal explanation.
2. Trace how molecular changes can scale up to organism, population, and ecosystem effects.
3. Use evidence from multiple course units to support a biological claim.
4. Identify feedback loops and tradeoffs in a biological system.

Overview

This capstone lab asks you to build a systems explanation instead of memorizing separate units. You will follow one scenario from molecule to ecosystem and show how the course ideas fit together.

Scenario

A drought changes plant growth in a local ecosystem. Some plant individuals carry a genetic variant that changes water-use efficiency. Herbivores, predators, decomposers, and nutrient cycling all respond over time.

Part 1: Scale Map

Complete the table with one effect at each scale.

Scale	What changes in the scenario?	Evidence or reasoning
Molecule / gene		
Cell		
Organism		
Population		

Scale	What changes in the scenario? Evidence or reasoning
Community	
Ecosystem	

Part 2: Causal Chain

1. Write a six-step causal chain from genetic variation to ecosystem-level change.

2. Where could natural selection act in this scenario? Identify the trait, environment, and fitness difference.

3. Where could feedback occur? Describe one reinforcing or balancing feedback loop.

Part 3: Evidence Check

Choose two course modules from Modules 01-16 and explain how each helps you reason about the drought scenario.

Module	Concept used	How it supports your explanation

Capstone Claim

4. Make one claim about how life works as a system. Support it with at least three ideas from different parts of BIOL-1.